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(1) Research Program/Project/Study Title: Growth Increment of Green Mussel Collected from the Offshore Mussel Culture in Pilar Bay, Capiz, Philippines	
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Academic Rank:	
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Program or Project has a component (if no proceed to 2)? ☐ Yes ☐ No

Component 1 Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	
Province:	
Region:	

(5) Classification Research: ☒ Basic ☐ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity	(8) Discipline/Field of Specialization

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(9) Rationale

Green mussel (*Perna viridis*) is one of the economically important bivalves in the Philippines, yielding 25,421 metric tons in 2019 (PSA, 2020). The term "tahong" refers to mussels, which are bivalve mollusks with an exterior covering consisting of a two-part hinged shell that encloses a soft-boiled invertebrate. The most common species in the Philippines is the *Perna viridis* (Salinas and Rosario, 2016). Native to the Indo-Pacific region, *P. viridis* is extensively dispersed from the Persian Gulf to the Philippines and Hong Kong (Siddall, 1980). They settle in areas of high water flow and can withstand turbid waters. They suck in water using an inhalant siphon and pump out water through an exhalant siphon. Their growth is influenced by fluctuations in temperature, light intensity, food concentration and composition, and water quality (Rejeki et. al, 2020). The habitats of green mussel are coastal regions at depths less than 10 meters (Power et al., 2004). The species colonize areas with fair fluctuations in salinity (Sivalinggam, 1977; Lee, 1988) usually occupying intertidal, subtidal, and estuarine environments. Salinity difference of the habitat of green mussel can cause distribution variation in population density (Huang et al, 1985).

Green mussels are considered as dioecious with no visible external dimorphism. However, sexes of species can be identified internally through colors of gonads and tissues. A sexually matured male have creamy-white colored gonads and whitish flesh while female have reddish gonads and orangey flesh (Cosling, 2003). Mussels is known to have self-regulate species that requires no commercial feed intake and minimal effort for growth (Ransangan&Soon, 2014). They usually grows 80-100 mm for one year, is a large and fast-growing bivalve (Power, et al 2014) with lifespan of approximately 2 - 3 years (Power et al., 2014).

The archipelago region of the Philippines is particularly well-suited for clam cultivation because its waters are home to spawning native species. The country's coastal waters are tropical, which offers ideal farming conditions for improved species development and reproduction. Cultivating mussels is a strategy worth pursuing to sustain the underprivileged fishermen because of their high value and inexpensive inputs. A few problems still hinder their development, though: mussel farms typically cover less than one hectare of land, and their output is only sold locally. However, the overcrowding of farms is a major issue in locations known for their shellfish, such as the province of Capiz. Numerous farm structures were neglected. One of the reasons for the decline in the commodities' production was the reclamation of mussel grounds because of siltation.

One of the initiatives of the Philippine government is to expand the production areas of mussel farming. Offshore area of Pilar Bay in Capiz is one of the selected areas for the possible expansion. There is a limited data on the suitability of offshore area of Pilar Bay for green shell aquaculture, although offshore mussel culture is already present. Thus, this study aims to determine the growth increment of Green Mussels cultured in the offshore culture system in Pilar Bay, Capiz.

(10) Objectives/Statement of the Problem

Generally, this study aims determine the growth increment of green mussels cultured in the offshore culture system in Pilar Bay, Capiz. Specifically, it aims to:

1. Determine the physico-chemical analysis of the water;
2. Determine the survival rate of green mussels in the offshore culture system ;
3. Determine the growth rate of the mussels in terms of weight, length, and thickness; and
4. Analyze the overall growth increment of the mussels in each sampling period

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Review of Related Literature

Green Mussel Industry

The term "tahong" refers to mussels, which are bivalve mollusks with an exterior covering consisting of a two-part hinged shell that encloses a soft-boiled invertebrate. Since mussels are an affordable source of protein and because mussel farming requires little capital investment, it offers additional revenue and a means of subsistence to fishermen in many coastal areas, making it an important part of the Philippines' aquaculture sector. In the Philippines, *P. viridis* is widely cultured for commercial purpose (Aypa, 1990). The aquaculture production of green mussel in 2018 reached to 26,302.77 mt valued at 9.8 M USD (PSA, 2019). In 2020, PSA reported a total of 19,228.96 MT production for the Philippines. Western Visayas led the mussel production with 8,534.16 MT, of which 7,945.70 MT came from the province of Capiz. Other high producing provinces are Cavite and Samar with 5,654.23 MT and 4,077.14 MT, respectively.

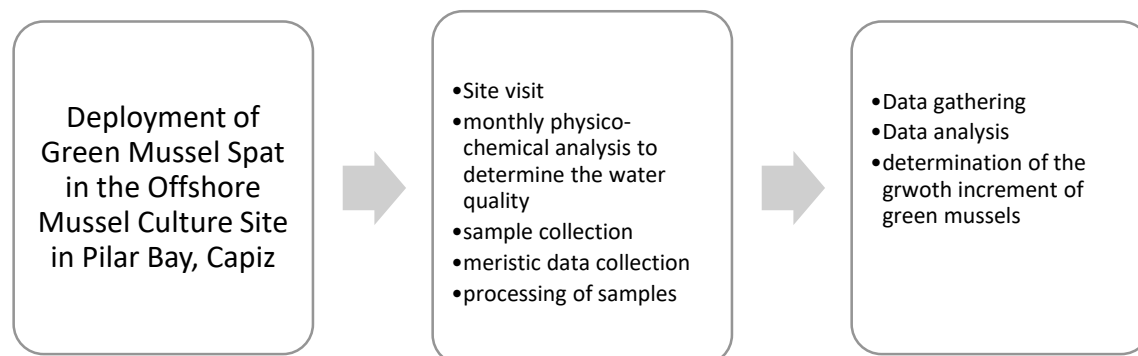
For the purpose of cultivating *P. viridis*, the natural environment—where seed stock is unstable—provides the supply of seed known as spats. A reliable source of seed for cultivated mussel species like *P. viridis* is essential to the shellfish industry's prosperity. With the aim to augment the seed supply and lessen the reliance of mussel farmers from getting their seeds in the wild, the Philippine government invested for a mussel hatchery in 2014 (Mero et al., 2019).

Biology of Mussels

Mussels are bivalve mollusks with an elongated and asymmetrical shell outline. They are known to exist in bays, coves and inlets where they grow best in brackish to salty waters, with salinity ranging from 27-35 ppt (varies between species). In an experiment conducted by A. Salinas (unpublished), it showed that Charru mussels can survive from 2 to 30 ppt but are already unable to develop byssus and dies when subjected at 35 ppt. They reach their sexual maturity during the first year and they spawn year-round having a peak period which ranges from May-June and September-October (Aypa, 1990). However, having an opportunistic type of gonadal cycle, this can vary depending on the fluctuations in the environment (e.g. temperature, food availability) which can trigger the spawning of bivalves.

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(11) Scientific Basis/Theoretical Framework



(13) Methodology

Study Site

Pilar Bay (Fig. 1) is situated at Capiz Philippines covering the three municipalities of the province; Municipality of Pontevedra, municipality of President Roxas, and Municipality of Pilar. These municipalities are considered as the main producers of green mussels in the province making Capiz as the top producer of green mussels in the Philippines as of 2022 (PSA, 2023).

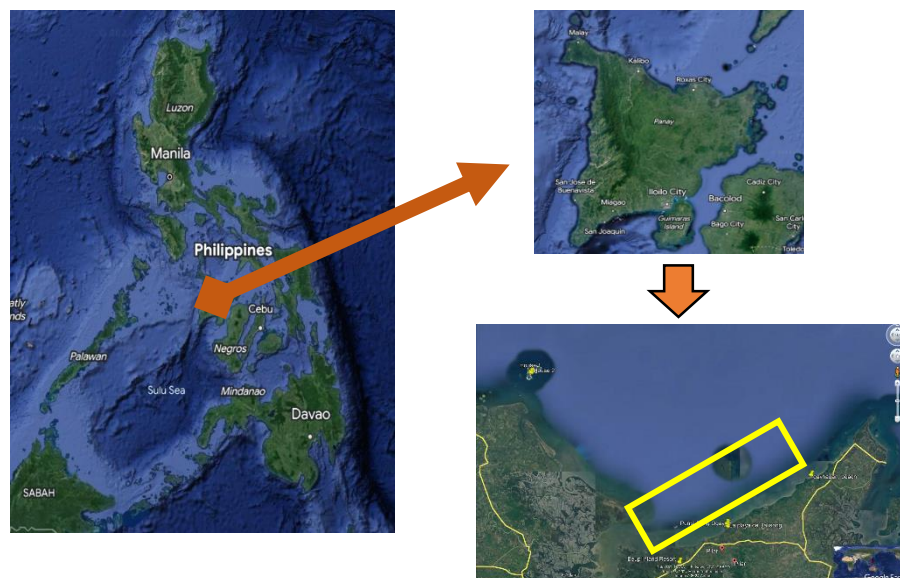


Figure 1. Sampling site of this study – Pilar Bay, Capiz, Philippines

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Physico-chemical Analysis

Monthly water parameters such as Dissolved Oxygen, Salinity, ORP, Conductivity, Temperature, Turbidity, and pH will be determined. This will help the researchers to determine the water quality and the effects of the fluctuation of the parameters to the growth of the samples

Deployment of the spat seeds of green mussel in the offshore mussel culture

Spat will be collected from Brgy. Madulano – one of the green mussel spats supplier not just in the province but also to the neighboring provinces. Spats will be placed in a net-sack (Fig. 2) to monitor the growth of the samples. One hundred spats will be deployed in each of the net sacks to get an accurate data.

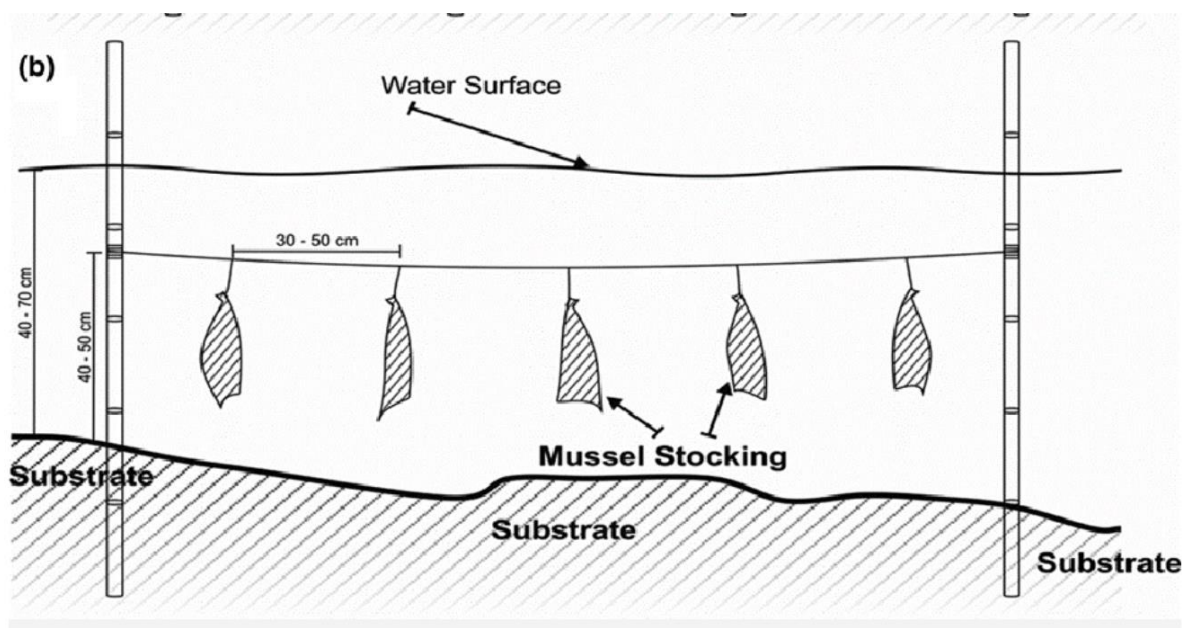


Figure 3. Model of the net-sack that will be employed in this study (Adapted from Rejeki et al., 2020)

Sample Collection

Freshly collected samples of green mussels will be collected from the offshore mussel culture in Pilar Bay, Capiz. A total of 5 sites will be determined during the pre-site visit. A total of 50 samples will be collected per sites during sampling collection. Meristic counts such as weight, length and thickness will be measure using a digital caliper. Photographs of the samples will also be taken in each of the samples.

Processing of green mussel samples

Individually, the meat samples will be extracted to determine the gender, wet meat weight, and dry met weight to determine the internal growth of the samples in each of the sampling months. The meat will be placed in a sealed container for data keeping.

Determination of the Survival Rate of the Samples

To determine the survival rate of the samples, the following formula will be employed (Muchlisin Iet al, 2016).

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$$SR = \frac{N_t - N_o}{N_o} \times 100\%$$

Where,

SR = Survival rate (%)

Nt = Number of shellfish at the final of research (sample)

No = number of shellfish at the start of research (sample)

Determination of the Growth Increment of the Samples

Green mussel growth and weight parameters including absolute length, absolute width, LPPM, and survival of green mussels will be analyze using the ANOVA test of variance prog, to see if there was a significant effect ($P < 0.05$) and continued with the Duncan further test to determine more clearly the differences.

Determination of the Absolute Length

The absolute length growth of green mussel shells was measured using the following formula (Soon et al., 2018).

$$L = L_2 - L_1$$

Where,

L = Absolute length (cm)

L2 = Final length (cm)

L1 = Initial length (cm)

Determination of the Absolute Weight

The weight growth of green mussels was measured using the following formula (Dewantoro, 2001).

$$W = W_t - W_o$$

Where,

W = Absolute weighted average (g)

Wt = Average weight of individuals at time – t

Wo = Average individual weight at the start of research

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(14) Major Activities/Work plan

WORKPLAN	M	J	J	A	S	O	N	D
PROPOSAL DRAFTING								
SITE VISIT								
MONTHLY WATER SAMPLING								
MONTHLY SAMPLE COLLECTION								
COLLECTION OF MERISTIC DATA								
DATA GATHERING								

(15) Expected Output

PUBLICATIONS

- 1 Research articles

PATENT/IP

- 1 IEC
- 1 Copyrighted materials

PRODUCT

-

PEOPLE SERVICES

-

PLACES AND PARTNETSHIP

-

POLICY

- Partnership with LGU for conservation policy

(15) Technology Transfer Activity Design

(16) Targets Beneficiaries

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(17) Personnel Requirement		
Name	Role (Program, Project, Study Leader)	Field of Specialization
Percent time devoted to the Program/ Project/ Study		
Program/Project Leader (in hours):		
Program/Project Members (in hours):		
(18) Budgetary Requirement		
(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, http://researchjournal.capsu.edu.ph/)		
Aypa, S. M. (1990). Mussel culture. In UNDP/FAO (RAS/90/002), Regional Seafarming Development and Demonstration Project (RAS/90/002), Selected Papers on Mollusc Culture. FAO. Cosling, E.M. 2003. Bivalve Molluscs: Biology, Ecology, and Culture. Malden, Oxford, Fishing News Books. http://dx.doi.org/10.1002/9780470995532 Lee SY. 1988. The reproductive cycle and sexuality of the green mussel <i>Perna viridis</i> (L.) (Bivalvia: Mytilacea) in Victoria Harbour, Hong Kong. Journal of Molluscan Studies. 54 (3): 317–323 Mero, F. F. C., Pedroso, F. L., Apines-Amar, M. J. S., Cadangin, J. F., Rendaje, D. C., Verde, C. S., Maquirang, J. R. H., & Piñosa, L. A. G. (2019). Influence of water management, photoperiod and aeration on growth, survival, and early spat settlement of the hatcheryreared green mussel, <i>Perna viridis</i>. International Aquatic Research. 11, 159–172. Power, A.J., Walker, R.L., Payne, K. and Hurley, D. 2004. First Occurrence of the Nonindigenous Green Mussel, <i>Perna viridis</i> in Coastal Georgia, United States. Journal of Shellfish Research, 23, 741-744. Philippines Statistics Authority (PSA). (2019). Fisheries Statistics of the Philippines 2016–2018, Volume 27. Quezon City: Philippines, Philippine Statistics Authority. 302p Rejeki S, Debrot A, den Brink A, Ariyati R, Widowati LL. 2020. Increased production in mussels (<i>Perna viridis</i>) using long culture and an economic comparison with stake culture in the north coast of Java, Indonesia. Aquac Res. 00:1-8 Salinas AD, Rosario WR. 2016. Biology and culture of the Philippine mussels. National Shellfish Development Program Manual No. 1. BFAR. Tacloban City. 20 pp. Siddall S. 980. A Clarification of the Genus <i>Perna</i> (Mytilidae). Bull Mar Sci. 30: 858–870. Soon, T.K., Azad, S.A. and Ransangan, J. 2014. Isolation and Characterization of Purple Non-Sulfur Bacteria, <i>Afifella marina</i>, Producing Large Amounts of Carotenoids from Mangrove Microhabitats. Journal of Microbiology and Biotechnology, 24, 1034-1043.		
(20) Capsule Curriculum Vitae (One-page brief C.V. only)		